Baye's.py

```python
def bayes_theorem(prior_h, likelihood_e_given_h, likelihood_e_given_not_h):
    """
    Calculate posterior probability using Bayes Theorem

    Args:
        prior_h: P(H) - Prior probability of hypothesis
        likelihood_e_given_h: P(E|H) - Likelihood of evidence given hypothesis
        likelihood_e_given_not_h: P(E|¬H) - Likelihood of evidence given not hypothesis

    Returns:
        posterior_h_given_e: P(H|E) - Posterior probability
    """

    # Calculate total probability of evidence P(E)
    p_e = (
        (likelihood_e_given_h * prior_h)
        + (likelihood_e_given_not_h * (1 - prior_h))
    )

    # Apply Bayes Theorem
    posterior = (likelihood_e_given_h * prior_h) / p_e

    return posterior, p_e


def display_result(
    hypothesis,
    prior,
    likelihood_yes,
    likelihood_no,
    posterior
):
    """
    Display results in a readable format
    """

    print("\n" + "=" * 60)
    print(f"Bayes Theorem Calculation: {hypothesis}")
    print("=" * 60)

    print(f"Prior Probability P(H): {prior:.4f}")
    print(f"Likelihood P(E|H): {likelihood_yes:.4f}")
    print(f"Likelihood P(E|¬H): {likelihood_no:.4f}")
    print(f"Prior Probability P(¬H): {1 - prior:.4f}")

    print("-" * 60)

    print(f"Posterior Probability P(H|E): {posterior:.4f}")
```

```
    print(f"Confidence Level: {posterior * 100:.2f}%")

    print("=" * 60)


# ============================================================
# Example 1: Medical Diagnosis
# ============================================================

print("\n" + "#" * 60)
print("EXAMPLE 1: Medical Diagnosis - Disease Detection")
print("#" * 60)

hypothesis_1 = "Patient has the disease"

prior_1 = 0.01
# 1% of population has disease

sensitivity = 0.95
# P(Test+|Disease) - True Positive Rate

specificity = 0.90
# P(Test-|¬Disease) - True Negative Rate

false_positive_rate = 1 - specificity
# P(Test+|¬Disease)

posterior_1, p_evidence_1 = bayes_theorem(
    prior_1,
    sensitivity,
    false_positive_rate
)

display_result(
    hypothesis_1,
    prior_1,
    sensitivity,
    false_positive_rate,
    posterior_1
)

print("\nInterpretation:")
print(
    "Even though the test is positive, the actual probability "
    "the patient"
)
print(
    f"has the disease is only {posterior_1 * 100:.2f}% (not 95%!)"
)
print(
```

```
    "This is due to the low prior probability of disease "
    "in the population."
)


# ============================================================
# Example 2: Spam Email Detection
# ============================================================

print("\n\n" + "#" * 60)
print("EXAMPLE 2: Spam Email Detection")
print("#" * 60)

hypothesis_2 = "Email is spam"

prior_2 = 0.30
# 30% of emails are spam

p_word_given_spam = 0.80
# 80% of spam contains the word "click"

p_word_given_not_spam = 0.05
# 5% of legitimate emails contain "click"

posterior_2, p_evidence_2 = bayes_theorem(
    prior_2,
    p_word_given_spam,
    p_word_given_not_spam
)

display_result(
    hypothesis_2,
    prior_2,
    p_word_given_spam,
    p_word_given_not_spam,
    posterior_2
)

print("\nInterpretation:")
print(
    f"If an email contains the word 'click', the probability "
    f"it is spam is {posterior_2 * 100:.2f}%"
)


# ============================================================
# Example 3: Weather Prediction
# ============================================================

print("\n\n" + "#" * 60)
```

```python
print("EXAMPLE 3: Weather Prediction - Rain Forecast")
print("#" * 60)

hypothesis_3 = "It will rain tomorrow"

prior_3 = 0.20
# 20% chance of rain (historical data)

p_cloudy_given_rain = 0.90
# 90% of rainy days are cloudy

p_cloudy_given_no_rain = 0.30
# 30% of non-rainy days are cloudy

posterior_3, p_evidence_3 = bayes_theorem(
    prior_3,
    p_cloudy_given_rain,
    p_cloudy_given_no_rain
)

display_result(
    hypothesis_3,
    prior_3,
    p_cloudy_given_rain,
    p_cloudy_given_no_rain,
    posterior_3
)

print("\nInterpretation:")
print(
    f"Given that tomorrow is cloudy, the probability "
    f"of rain is {posterior_3 * 100:.2f}%"
)


# ============================================================
# Comparison Table
# ============================================================

print("\n\n" + "#" * 60)
print("SUMMARY: Posterior Probabilities for All Examples")
print("#" * 60)

print(
    f"{'Example':<30} {'Prior':<12} {'Posterior':<12}"
)

print("-" * 60)

print(
```

```
    f"{'Disease Detection':<30} "
    f"{prior_1:<12.4f} "
    f"{posterior_1:<12.4f}"
)

print(
    f"{'Spam Detection':<30} "
    f"{prior_2:<12.4f} "
    f"{posterior_2:<12.4f}"
)

print(
    f"{'Rain Prediction':<30} "
    f"{prior_3:<12.4f} "
    f"{posterior_3:<12.4f}"
)

print("=" * 60)
```

```
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PS C:\Users\Student.DESKTOP-R275QHQ\python folder 1> python -u "c:\Users\Student.DESKTOP-R275QHQ\python folder 1\Baye's.py"

############################################################
EXAMPLE 1: Medical Diagnosis - Disease Detection
############################################################

============================================================
Bayes Theorem Calculation: Patient has the disease
============================================================
Prior Probability P(H): 0.0100
Likelihood P(E|H): 0.9500
Likelihood P(E|¬H): 0.1000
Prior Probability P(¬H): 0.9900
------------------------------------------------------------
Posterior Probability P(H|E): 0.0876
Confidence Level: 8.76%
============================================================

Interpretation:
Even though the test is positive, the actual probability the patient
has the disease is only 8.76% (not 95%!)
This is due to the low prior probability of disease in the population.


############################################################
EXAMPLE 2: Spam Email Detection
############################################################

============================================================
Bayes Theorem Calculation: Email is spam
============================================================
Prior Probability P(H): 0.3000
Likelihood P(E|H): 0.8000
Likelihood P(E|¬H): 0.0500
Prior Probability P(¬H): 0.7000
------------------------------------------------------------
Posterior Probability P(H|E): 0.8727
Confidence Level: 87.27%
============================================================
```

```
Interpretation:
If an email contains the word 'click', the probability it is spam is 87.27%


############################################################
EXAMPLE 3: Weather Prediction - Rain Forecast
############################################################

============================================================
Bayes Theorem Calculation: It will rain tomorrow
============================================================
Prior Probability P(H): 0.2000
Likelihood P(E|H): 0.9000
Likelihood P(E|¬H): 0.3000
Prior Probability P(¬H): 0.8000
------------------------------------------------------------
Posterior Probability P(H|E): 0.4286
Confidence Level: 42.86%
============================================================

Interpretation:
Given that tomorrow is cloudy, the probability of rain is 42.86%


############################################################
SUMMARY: Posterior Probabilities for All Examples
############################################################
Example                        Prior        Posterior
------------------------------------------------------------
Disease Detection              0.0100       0.0876
Spam Detection                 0.3000       0.8727
Rain Prediction                0.2000       0.4286
============================================================
```